Giovanni Tauro

R&D Engineer, PhD Candidate| Rehabilitation Robotics & IoT



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LinkedIn GitHub Scholar Portfolio GrabCAD

R&D Engineer and PhD Candidate with 5+ years of applied research at CNR-STIIMA in rehabilitation robotics, wearable sensing, and IoT systems. Patent co-inventor and co-founder of Rehabilia Technologies, turning research prototypes into products validated with patients. I bridge academic research and industrial R&D, from experimental design to technology transfer.

Work Experience

R&D Product Engineer, STIIMA, Lecco
Feb 2021 – Present

- Developed sensor-based rehabilitation and assistive technology systems, translating clinical requirements and user needs into functional prototypes for experimental validation .
- Defined the architecture of multimodal IoT feedback platforms, integrating wearable sensing, indoor localization, BLE communication, and adaptive audio, visual, and haptic feedback.
- Delivered 3+ functional prototypes across embedded electronics, wearable devices, and interactive systems, supporting projects from concept development to ISO compliance technology delivery.
- Designed communication infrastructures and evaluation protocols for digital health applications, assessing usability, system performance, and user interaction through experimental testing.
- **Head of Product (Co-Founder)**, Rehabilia Technologies s.r.l. (May 2023 – Present), applying this R&D expertise to lead product development as Head of Product:
  - Led end-to-end product development of rehabilitation technologies, from user research and problem discovery to validation and product delivery.
  - Defined product requirements and priorities by translating user needs, business goals, and stakeholder feedback into actionable roadmaps.
  - Collaborated with engineering, design, and clinical stakeholders to align product strategy and deliver iterative product improvements.
  - Validated product decisions through usability testing, interviews, and data-driven loops.

`Embedded Systems • IoT • ESP32 • Arduino • PlatformIO • VS • BLE GATT • ROS2 • C++ • C# • Python • Prototyping • 3DCAD • FDM • Material Selection • UX Research • Product Development • Stakeholder Communication • ASANA • Cross-functional Teamwork • Project Management • Public Speaker`

Freelancer, Florence
Jan 2016 – Sep 2017, collaboration with DIMA3D

- Delivered end-to-end reverse engineering workflows, converting 3D scan data into validated parametric CAD models and production-ready assets.

`CAD • Reverse Engineering • Geomagic • SolidWorks • 3D Scanning • Product Development • Technical Documentation • Manufacturing Preparation`

Selected Projects



GIT_1
GIT_2
GIT_3

OmniTrack: IoT-based Multimodal Navigation System /Frontiers

- Built end-to-end hardware/software prototypes, validated with 16 blind participants, integrating IoT-oriented sensing, processing, and user feedback into a complete system.
- Applied real-time data processing and system integration principles to transform physical-world signals into actionable information.

`Python` • `Data Anlysis` • `C++` • `JS` • `Dashboard` • `ESP32` • `ROS2` • `BLE GATT` • `Sensors` • `I2C` • `Linux`



WEB

PhiCube: reconfigurable bilateral robotic device for neurorehabilitation /Frontiers

- Designed a modular rehabilitation device combining UX research, mechanical design, embedded systems, and multimodal interaction.
- Validated with 19 children (SUS 85.8/100); 134 patients in two on-going multicentric studies.

`UX` • `Interaction Design` • `CAD` • `Prototyping` • `Assembly` • `Usability`

RehaBalloon: Interactive Platform



GIT

- Prototyped a low-cost rehabilitation device in 24 hours during the LMU Munich "Hacking with Hardware" challenge, integrating pressure sensing, visual feedback, and embedded electronics.
- Implemented a TensorFlow Lite model on Arduino embedded machine learning classifier exercise patterns and record rehabilitation activity metrics in real time.

`Seeed Studio` • `Sensors` • `C++` • `TensorFlow` • `Machine Learning`

Education

2023–27 **PhD, Mechanics and Advanced Engineering Sciences** *Bologna University, IT*

PhD Candidate, expected 2027. Topic: IoT-based Multimodal Navigation System – OmniTrack
• Research Mobility, Utrecht University, NLD: Sensory navigation and experimental psychology.
• Research Mobility, Toyo University, JP: Feedback algorithm and behaviour analysis.

2017–20 **MSc, Design & Engineering** *Politecnico di Milano, IT*

Topic: UX development of an ultrasonic dental device: from concept to validation: WashB
• Research Mobility, Chiba University, JP: UX research studies for Fujitsu and Panasonic.

2012–15 **BSc, Industrial Design** *Florence University, IT*

Topics: Foundation in research methodologies, development, and critical thinking.

Certifications

2026 **Claude 101 Badge — Claude Code Badge — Claude** *Anthropic, Online*

2024 **.ppt — Robotics and Allied Technologies for Rehab** *MUR, Rome, Italy*

Patent

2023 Malosio, M., Nicora, M. L., Redaelli, D., Tauro, G. WO2023214349A1 *Device for bilateral rehabilitation*.

Languages

Italian: Native English: Advanced (IELTS 2017) French: Basic (A2)

Lecco, October 9, 2026

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